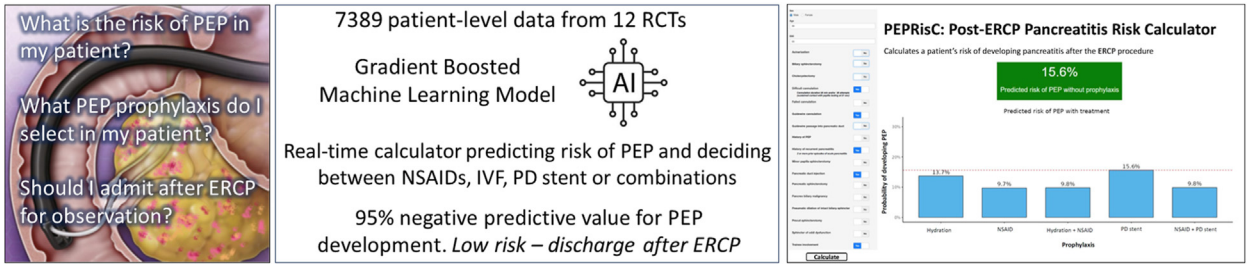
# Development and validation of a machine learning–based, point-of-care risk calculator for post-ERCP pancreatitis and prophylaxis selection

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## GRAPHICAL ABSTRACT




**Background and Aims:** A robust model of post-ERCP pancreatitis (PEP) risk is not currently available. We aimed to develop a machine learning–based tool for PEP risk prediction to aid in clinical decision making related to periprocedural prophylaxis selection and postprocedural monitoring.

**Methods:** Feature selection, model training, and validation were performed using patient-level data from 12 randomized controlled trials. A gradient-boosted machine (GBM) model was trained to estimate PEP risk, and the performance of the resulting model was evaluated using the area under the receiver operating curve (AUC) with 5-fold cross-validation. A web-based clinical decision-making tool was created, and a prospective pilot study was performed using data from ERCPs performed at the Johns Hopkins Hospital over a 1-month period.

**Results:** A total of 7389 patients were included in the GBM with an 8.6% rate of PEP. The model was trained on 20 PEP risk factors and 5 prophylactic interventions (rectal nonsteroidal anti-inflammatory drugs [NSAIDs], aggressive hydration, combined rectal NSAIDs and aggressive hydration, pancreatic duct stenting, and combined rectal NSAIDs and pancreatic duct stenting). The resulting GBM model had an AUC of 0.70 (65% specificity, 65% sensitivity, 95% negative predictive value, and 15% positive predictive value). A total of 135 patients were included in the prospective pilot study, resulting in an AUC of 0.74.

**Conclusions:** This study demonstrates the feasibility and utility of a novel machine learning–based PEP risk estimation tool with high negative predictive value to aid in prophylaxis selection and identify patients at low risk who may not require extended postprocedure monitoring. (Gastrointest Endosc 2025;101:129-38.)

*(footnotes appear on last page of article)*

Post-ERCP pancreatitis (PEP) is the most common adverse event associated with ERCP, with a reported incidence between 7% and 10% in average-risk populations and up to 15% in high-risk patient populations as well as significant associated morbidity and cost.[1,2] Although many retrospective and prospective studies have evaluated patient and procedural risk factors for the development of PEP in the past, there are no validated clinical decision-making tools to predict the risk of developing PEP.[3-14] The recently published guidelines on the prevention of PEP highlight the need for accurate, personalized PEP risk assessment using demographic and periprocedural parameters.[15]

Several prognostic scoring tools to stratify patients based on their risk of developing PEP have been developed in the past, but these are rarely implemented in clinical practice.[16-21] Prior models have relied on multivariable linear regression models, which cannot account for the nonlinear, interdependent relationships among the multiple PEP risk factors.[22] Second, these scoring tools have relied on a few patients and procedural variables with limited statistical validation. Furthermore, none of these tools have been capable of predicting the reduction in PEP risk associated with well-characterized PEP prophylaxis strategies such as rectal nonsteroidal anti-inflammatory drugs (NSAIDs), aggressive hydration, and prophylactic pancreatic duct (PD) stenting.[23]

Machine learning methods of computational modeling, such as random forest algorithms and gradient boosting, are designed to develop high-level pattern recognition by integrating large numbers of covariates that can be further improved through data experience.[24] The application of these machine learning techniques to predict the risk of PEP and potential benefit from prophylactic interventions, however, have not yet been explored. In this study, we aimed to develop and validate a random forest gradient-boosted machine (GBM) learning model to predict the risk of PEP with and without prophylactic interventions. We have further developed this model into a point-of-care decision-making tool to aid clinicians in PEP prophylaxis selection and help identify patients at low risk of developing PEP who do not require extended postprocedural monitoring in the hospital.

## MATERIALS AND METHODS

### Data integration

Individual patient-level data from 12 multicenter randomized controlled trials (Table 1) were combined to construct and train a prediction model. A standard set of 20 patient and procedural risk factors for PEP (Table 2) and 5 prophylactic interventions (aggressive hydration, rectal NSAIDs, the combination of aggressive hydration and rectal NSAIDs, PD stenting, and the combination of rectal NSAIDs and PD stenting) were considered based on the availability of data from these trials and the relevance to current practice patterns. For uniformity, any form of aggressive intravenous therapy hydration (normal saline, lactated Ringer's solution) was considered because no major differences across these were noted in our prior research.[23] Rectal indomethacin 100 mg and rectal diclofenac 100 mg were considered equivalent and labeled as rectal NSAIDs. The timing of NSAID administration (preprocedure/intraprocedure/postprocedure) was not differentially analyzed. Patient-level data were available from the included studies; therefore, patients receiving PD stents alone or in combinations were identified, even if they were not randomized for this intervention as such. Patients receiving placebos or no intervention were identified as a control group. For each continuous variable, we calculated the variable mean and the incidence of PEP among those with a value greater than the mean. For each binary risk factor, we display the variable mean (equivalent to the proportion of patients who possess that risk factor) and the incidence of PEP among those who had the risk factor.

All 12 studies used the revised consensus criteria definition of acute pancreatitis.[37] Five of these studies included patients at any risk of PEP, and the other 7 studies included patients at moderate-to-high or high risk for the development of PEP. Common inclusion criteria for these studies with enriched patient populations included clinical suspicion of sphincter of Oddi dysfunction, history of PEP, pancreatic sphincterotomy, precut sphincterotomy, more than 8 cannulation attempts (as determined by the endoscopist), pneumatic dilation of an intact biliary sphincter, or ampullectomy. Patients were also eligible if they met 2 or more of the following minor criteria: <50 years old and female, history of recurrent pancreatitis (>2 episodes), 3 or more contrast injections into the PD with at least 1 injection to the tail of the pancreas, pancreatic acinarization, or PD brush cytology. Common exclusion criteria across all 12 studies included unwillingness or inability to consent for the study; age of <18 years; pregnancy; breastfeeding; heart failure or respiratory insufficiency; and presence of standard contraindications to ERCP (eg, uncontrolled coagulopathy or hemodynamic instability), active or recent (within 4 weeks) GI hemorrhage, acute pancreatitis (serum amylase and/or lipase peak of >3 times the upper limit of normal) within 72 hours, known chronic calcific pancreatitis, or pancreatic head mass. In the 9 studies evaluating the effects of NSAIDs on PEP, patients with contraindications or allergies to NSAIDs, renal impairment, or recent NSAID use within 72 hours were also excluded.

### Statistical model

The statistical analysis was performed using R 4.0.0 (R Foundation for Statistical Computing, Vienna, Austria).

**Feature selection and imputation.** A set of risk factors common to all 12 studies was constructed, and those with rare occurrences (<25 patients) were excluded from the model. Missing values in the remaining risk factors were imputed using median imputation and *k*-nearest neighbors. Samples in studies 5 and 12 with missing values

**TABLE 1. Studies and baseline patient characteristics**

| Study number | Study authors | Number of patients | PEP incidence, n (%) | Age, y, mean ± SD | Sex, % male | Prophylactic interventions |
|---|---|---|---|---|---|---|
| 1 | Buxbaum et al[25] | 62 | 4 (6.5) | 43.8 ± 15.0 | 48.4 | Aggressive hydration, rectal NSAIDs, or control |
| 2 | Choi et al[26] | 510 | 36 (7.1) | 59.8 ± 11.6 | 54.5 | Aggressive hydration, rectal NSAIDs, PD stenting, or control |
| 3 | Elmunzer et al[27] | 602 | 79 (13.1) | 45.3 ± 13.3 | 20.9 | Rectal NSAIDs, PD stenting, or control |
| 4 | Fogel et al[28] | 1037 | 141 (13.6) | 49.8 ± 15.1 | 21.6 | Rectal NSAIDs |
| 5 | Kamal et al[29] | 959 | 63 (6.6) | 52.3 ± 15.0 | 42.5 | Rectal NSAIDs |
| 6 | Lai et al[30] | 162 | 9 (5.6) | 60.0 ± 16.3 | 63.0 | Rectal NSAIDs |
| 7 | Levenick et al[31] | 449 | 27 (6.0) | 64.6 ± 15.0 | 47.4 | Rectal NSAIDs, PD stenting, or control |
| 8 | Lua et al[32] | 144 | 11 (7.6) | 49.9 ± 17.1 | 41.0 | Rectal NSAIDs, PD stenting, or control |
| 9 | Luo et al[33] | 2292 | 134 (5.9) | 60.7 ± 15.7 | 47.8 | Rectal NSAIDs, PD stenting, or control |
| 10 | Mok et al[34] | 192 | 28 (14.6) | 60.3 ± 17.1 | 37.5 | Rectal NSAIDs, aggressive hydration, or PD stenting |
| 11 | Phillip et al[35] | 167 | 31 (18.6) | 60.7 ± 14.5 | 42.5 | PD stenting or control |
| 12 | Sperna Weiland et al[36] | 813 | 69 (8.5) | 57.2 ± 16.4 | 40.7 | Aggressive hydration or rectal NSAIDs plus aggressive hydration |

The number of patients, PEP incidence, mean age, sex (percent male), randomization protocol, and PEP prophylactic therapies studied are listed for each trial included in the study.
*NSAID*, Nonsteroidal anti-inflammatory drug; *PD*, pancreatic duct; *PEP*, post-ERCP pancreatitis; *SD*, standard deviation.

**TABLE 2. Risk factors included in the GBM model ranked by variable importance**

| Variable | Variable mean | PEP incidence | Importance |
|---|---|---|---|
| BMI,* $kg/m^2$ | 24.820 | 0.098 | 100.0 |
| Age,* y | 56.227 | 0.069 | 72.4 |
| Guidewire passage into pancreatic duct | 0.357 | 0.151 | 54.7 |
| >2 Guidewire passages into pancreatic duct | 0.099 | 0.136 | 26.5 |
| >2 Pancreatic duct injections | 0.066 | 0.188 | 22.6 |
| Pancreatic duct injection | 0.206 | 0.159 | 15.0 |
| Cholecystectomy | 0.266 | 0.080 | 14.5 |
| Minor papilla sphincterotomy | 0.123 | 0.384 | 13.5 |
| Difficult cannulation | 0.292 | 0.114 | 12.5 |
| History of PEP | 0.043 | 0.213 | 11.3 |
| Pneumatic dilation of intact biliary sphincter | 0.018 | 0.135 | 10.9 |
| Female sex | 0.407 | 0.064 | 8.0 |
| Failed cannulation | 0.049 | 0.102 | 7.9 |
| Pancreatic sphincterotomy | 0.154 | 0.151 | 6.6 |
| Acinarization | 0.032 | 0.207 | 5.5 |
| Biliary sphincterotomy | 0.736 | 0.086 | 4.2 |
| Guidewire cannulation | 0.935 | 0.071 | 3.0 |
| History of recurrent pancreatitis | 0.101 | 0.152 | 2.8 |
| Precut sphincterotomy | 0.120 | 0.114 | 2.5 |
| Sphincter of Oddi dysfunction | 0.177 | 0.139 | 2.4 |
| Pancreaticobiliary malignancy | 0.088 | 0.082 | 1.8 |
| Trainee involvement | 0.261 | 0.093 | 0.0 |

*GBM*, Gradient-boosted machine; *PEP*, post-ERCP pancreatitis.
*For each continuous variable, we display the variable mean, the incidence of PEP among those with a value greater than the mean, and the variable importance under the trained GBM. For each binary risk factor, we display the variable mean (equivalent to the proportion of patients for whom the risk factor occurred), the incidence of PEP among those who had the risk factor, and the variable importance under the trained GBM. The variable importance is found by calculating the average empirical improvement from splitting on the given risk factor.

were first imputed using the median because these studies had lower rates of missing data and were only missing values in a few binary variables (Table 1). The rest of the missing values in other samples were then imputed as the mean of the 10 nearest neighbors from studies 5 and 12 using the caret package. In all steps, values were imputed using samples from the training set only (80% of the patients under 5-fold cross-validation).

**Prediction model.** A GBM model was fitted using the caret R package and tuned using the Kappa statistic. Since the model is trained on patient-level data, where one observation represents one patient, studies with larger sample sizes are implicitly weighted more in the model. The number of trees and the interaction depth of the tree were tuned under 5-fold cross-validation over a grid of 50, 100, 150, 500, and 1000 trees and a depth of 1, 2, 3, and 4, while the learning rate was held constant at 0.1 and the minimum number of training set samples in a node to commence splitting was held constant at 10. The model trained over the entire dataset had tuning parameters of 150 trees and an interaction depth of 3. To evaluate the performance of the model, a 5-fold cross-validation was performed. Results were additionally compared to a ridge logistic regression model and a random forest classifier. For ridge logistic regression, the lambda tuning parameter was tuned over a grid of 0, 0.0001, and 0.1. For the random forest classifier, the number of variables randomly sampled as candidates at each split was tuned over a grid of 2, 13, and 25.

**Variable importance.** To assess the marginal importance of each risk factor, we calculated the incidence of PEP for each variable. For continuous variables like age, we calculated the incidence of PEP among those with an age greater than the mean age in the dataset. For binary variables like trainee involvement, we calculated the incidence of PEP among patients with trainee involvement. In addition, we ranked the variables by the mean decrease in entropy or logistic loss when a tree is split on that variable.[38] This represents the contribution of each risk factor toward the predictive value of the model and is an overall metric for the variable importance.

**Prophylaxis effects.** The algorithm determines what prophylaxis the patient will benefit from, in addition to providing a percentage PEP risk probability. We considered the effects of 5 prophylaxis regimens—aggressive hydration, rectal NSAIDs, the combination of aggressive hydration and rectal NSAIDs, PD stenting, and the combination of rectal NSAIDs and PD stenting. To obtain unbiased estimates, each treatment effect was estimated using only the subset of patients for whom the treatment was randomized in the included studies. The exception is prophylactic PD stenting, which was not a randomized treatment in most studies and was used only when clinically indicated. Given the limitations of the available nonrandomized data, the prophylactic effect of PD stenting was therefore estimated using all studies that administered PD stents to any patients. The prophylactic effect of the rectal NSAID and PD stenting combination was estimated using the subset of patients for whom the use of rectal NSAIDs was randomized in the corresponding studies.

To estimate the treatment effect of treatment $t$ and a new patient $i$, the prediction score $p_i$ is first estimated using the full model, with values for all treatment options set to 0. Then a second model $M_t$ is trained on the subset of patients from studies that randomized treatment $t$. Using this model, 2 new prediction scores are obtained for patient $i$: (1) $p_{it_0}$, which is the prediction score from $M_t$ when patient $i$ did not receive treatment $t$, and (2) $p_{it_1}$, which is the prediction score from $M_t$ when patient $i$ received treatment $t$. The ratio of these 2 values, $a_{it} = \frac{p_{it_1}}{p_{it_0}}$, is the adjustment factor. The prediction score for patient $i$ with treatment $t$ is estimated to be $p_{it} = p_i * a_{it}$, and the treatment effect is, therefore, the difference between the 2 prediction scores $p_{it} - p_i$. For the aggressive hydration and rectal NSAID combination prophylaxis, this procedure was performed twice. Because no studies randomized patients to the combination treatment, 2 adjustment factors were computed, 1 for indomethacin ($a_{it_{\text{indo}}}$) and another for aggressive hydration ($a_{it_{\text{ah}}}$). The overall adjustment factor for the indomethacin and aggressive hydration combination is taken as the product $a_{it_{\text{indo}}} * a_{it_{\text{ah}}}$.

**Web-based application development.** A PEP risk calculator and decision-making tool were developed using the R Shiny package (RStudio, Boston, Mass, USA). This web-based application was designed to serve as a user-friendly interface for the GBM model, allowing the user to input patient-specific demographic and procedural variables and receive an output containing that specific patient's predicted risk of PEP at baseline (no intervention) and under each of the included prophylactic therapeutic models.

**Prospective feasibility study.** Procedural data from all ERCPs performed at the Johns Hopkins Hospital in the month of September 2022 were collected prospectively and input into the PEP risk calculator to assess the baseline risk of PEP (no prophylactic intervention) and under each of the 5 prophylactic therapeutic models. Patient charts were monitored for the development of pancreatitis within 5 days of the procedure using the revised consensus criteria definition of acute pancreatitis.[37] The prospective predictive value of the model was assessed using a receiver operating curve generated using Prism 10 (GraphPad Software Inc., La Jolla, Calif, USA). Prediction of relative risk reduction with each therapeutic modality was estimated as the predicted risk of PEP with prophylaxis minus the baseline risk of PEP divided by the baseline risk of PEP.

## RESULTS

### Study characteristics

A total of 7389 patients from 12 trials were included, among whom 632 (8.6%) developed PEP. In total, there

were 5060 (68.4%) patients who received rectal NSAIDs, 665 (9.0%) who received aggressive hydration, 365 (4.9%) who received a combination of aggressive hydration and rectal NSAIDs, 1645 (22.3%) who received PD stenting, and 1134 (15.3%) who received a combination of rectal NSAID and PD stenting, whereas the remaining 1588 (21.5%) patients did not receive any prophylaxis.

### Variable selection and importance

Twenty risk factors, reported in prior literature and common to all 12 included trials, were identified and are summarized in Table 2. Among the risk factors, PD brush cytology and pancreas divisum were excluded from the model because of their rare occurrence (<25 patients out of 7389 total patients), and the classification of sphincter of Oddi dysfunction type was excluded because of inconsistent reporting across the included trials. Of the remaining risk factors, importance was calculated based on the decrease in entropy in the model afforded by decision trees splitting on that variable. This analysis (Table 2) showed that patient age and body mass index were the most important patient and demographic factors, and the most important procedural factors were those involving direct PD instrumentation, including PD guidewire cannulation and contrast injection. Notably, the risk factors identified as having the greatest impact on the predictive capacity of the model were not those associated with the highest incidence of PEP. These included minor papillary sphincterotomy (38.4% PEP positive), history of PEP (21.3% PEP positive), and pancreatic acinarization (20.7% PEP positive).

### Prediction performance

Figure 1 displays the predictive performance of the model, represented by the receiver operating characteristic curve and calibration plot. The area under the curve (AUC) under 5-fold cross-validation was 0.70 for all patients. Using a predicted risk cutoff chosen by maximizing the Youden's index, our model achieved an accuracy of 65%, a negative predictive value of 95%, a positive predictive value of 15%, a specificity of 65%, and a sensitivity of 65%. This was comparable to or better than the performance metrics achieved by a multivariable ridge logistic regression and a random forest classifier (Table 3).

### Risk estimation calculator

To aid with decision making, we packaged our model into a web-based application that serves as a risk estimation calculator and prediction tool. Figure 2 demonstrates this application and user interface using a sample patient. This allows the endoscopist to input patient and procedural risk factor data and receive the predicted risk of that patient developing PEP without intervention and under each of the included prophylactic interventions in real time. As shown in Figure 2, the endoscopist can consider the appropriate PEP prophylaxis for the patient and decide on extended postprocedure observation versus discharge.

### Prospective feasibility study

Data were collected from 135 ERCP cases performed at the Johns Hopkins Hospital, including 14 cases of PEP (10.4%). Figure 3A shows the receiver operating characteristic for the PEP risk estimation calculator in this prospective cohort, which generated an AUC of 0.74. The predicted decrease in PEP risk with each prophylactic therapy is shown in Figure 3B.

## DISCUSSION

Early assessment of risk for PEP before ERCP is essential to determine the need for prophylaxis and postprocedure monitoring. To aid in the rapid point-of-care prediction of PEP risk, we have developed a gradient-boosted, random forest machine learning model using patient-level data from 12 large randomized controlled clinical trials. This model predicts the risk of PEP in a given patient with a cross-validated AUC of 0.70 and estimates the reduction in PEP risk across 5 prophylactic regimens. With its high negative predictive value, our model accurately identifies patients at low risk of developing PEP. This model can potentially be used to identify patients who may not require extended postprocedure monitoring or extensive PEP prophylaxis based on the endoscopist's approach.

The use case in Figure 2 outlines a prototypical scenario in which a young female patient with risk factors for PEP undergoes an ERCP with a difficult cannulation but no wire cannulation of the PD. Using our PEP risk prediction model, the endoscopist could determine, using this patient's demographic and procedural risk factors, whether the potential benefits of PD cannulation and PD stent placement outweigh the potential risks. In this patient's case, we see that PD stent placement increases the risk of PEP as compared to hydration and NSAID prophylaxis. As such, the endoscopist using this prediction model could avoid an additional procedural aspect that would confer no additional benefit to the patient and may potentially increase the risk of PEP. At the conclusion of the procedure, the endoscopy team could use this same data to assess, based on the absolute risk of PEP after the prophylactic interventions performed, whether a given patient can safely be discharged from the postanesthesia care unit or if their risk of PEP is high enough that they would benefit from prolonged postprocedural observation in the hospital.

Models for predicting and stratifying PEP risk have been proposed based on a variety of demographic and procedural factors. Friedland et al[16] in 2002 used logistic regression to generate risk scores based on 4 factors—pain during the procedure, PD cannulation, history of PEP, and the number of cannulation attempts. Since then, several models using a comparable methodology and varied combinations of risk

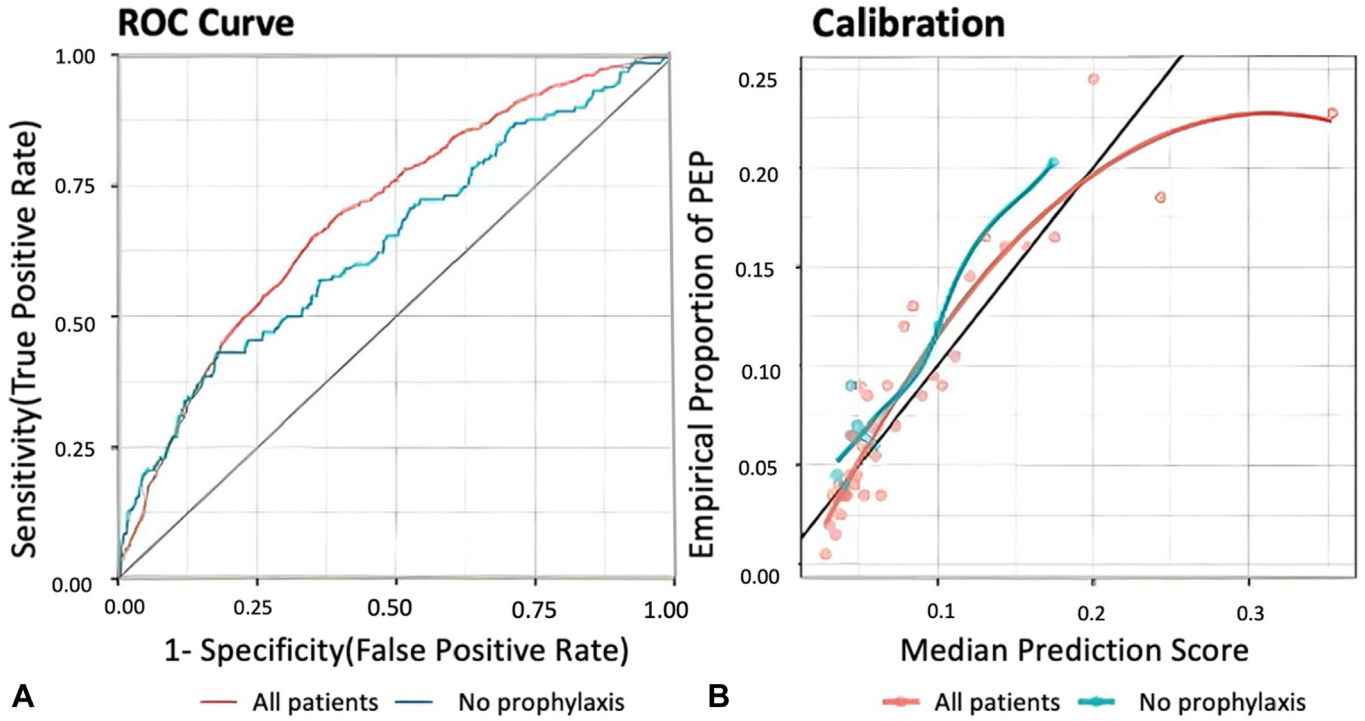



**Figure 1.** Prediction performance of the gradient-boosted machine model. **A,** We obtain an area under the curve of 0.70 under 5-fold cross-validation for all patients *(red curve)* and of 0.64 for the subset of patients without prophylaxis *(blue curve)*. **B,** Patients are ranked by their predicted PEP risk and grouped per 200 patients. For each group, we plot the median prediction score from our model (*x*-axis) against the proportion of patients who developed PEP (*y*-axis). Points lie along the $y = x$ line *(black)* for a well-calibrated model. *PEP*, Post-ERCP pancreatitis; *ROC*, receiver operating characteristic.

**TABLE 3. Prediction performance metrics**

| Model | AUC | Accuracy | NPV | PPV | Specificity | Sensitivity |
|---|---|---|---|---|---|---|
| All patients | | | | | | |
| Gradient-boosted machine | 0.70 | 0.65 | 0.95 | 0.15 | 0.65 | 0.65 |
| Ridge regression | 0.70 | 0.58 | 0.96 | 0.14 | 0.56 | 0.73 |
| Random forest | 0.65 | 0.56 | 0.95 | 0.12 | 0.55 | 0.68 |
| No prophylaxis | | | | | | |
| Gradient-boosted machine | 0.64 | 0.79 | 0.94 | 0.18 | 0.82 | 0.43 |
| Ridge regression | 0.67 | 0.81 | 0.94 | 0.20 | 0.85 | 0.42 |
| Random forest | 0.63 | 0.56 | 0.94 | 0.12 | 0.55 | 0.64 |

The performance of each model under 5-fold cross-validation is shown. The predicted probability cutoff was chosen to maximize the Youden index.
*AUC*, Area under the curve; *NPV*, negative predictive value; *PPV*, positive predictive value.

factors have been presented, although all have been limited by their use of retrospective and nonrandomized data, as well as the incorporation of small numbers of risk factors and limited ability to account for the interaction among risk factors.[17-19] A recent scoring system developed by our group applied logistic regression to randomized controlled trial data and used classification and regression tree analysis to designate risk score cutoffs.[20] Despite a technical advancement over previous risk scores, this model performed poorly in validation cohorts because of interactions among risk factors.

The use of a web-based application for predicting PEP risk has been previously proposed by Rodrigues-Pinto et al.[19] We believe that our implementation of this idea has several key advantages in addition to the use of a more statistically robust methodology and the inclusion of additional PEP-specific risk factors. Our application is the first, to our knowledge, to integrate prophylactic therapies for PEP prevention and confer a predicted reduction in the PEP risk for an individual patient. These features will allow clinicians to rapidly make patient-specific decisions at the point of care through the real-time prediction of PEP risk as well as expected treatment benefits from common prophylactic strategies.

The key methodologic advantage of our GBM forest model is the ability to account for interdependencies and

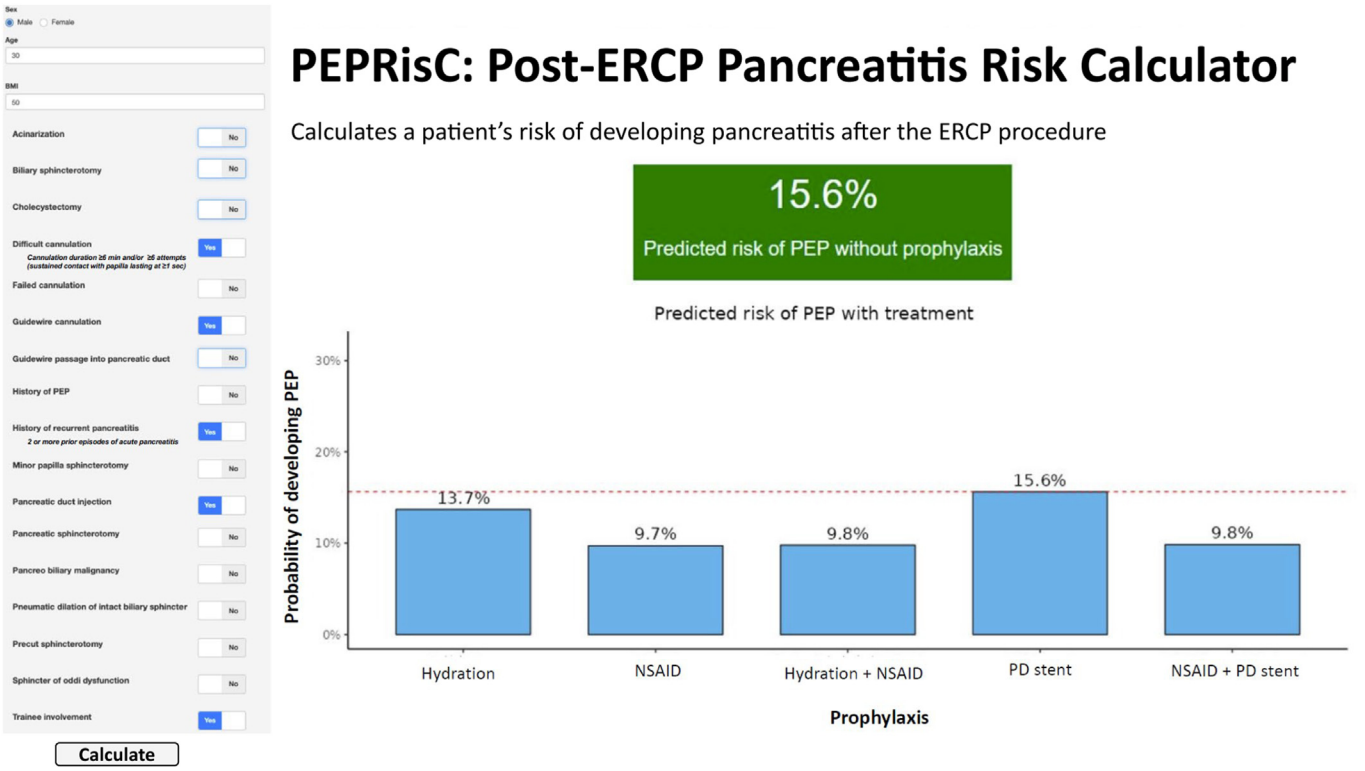



**Figure 2.** Web-based PEP risk calculator and decision-making tool. An application was created using an R-based software suite to serve as a clinical decision-making aid. The user interface include inputs for each variable in the model *(left)* with the output displayed on the right, including the baseline predicted risk of PEP in green *(top right)* as well as the predicted risk of PEP associated with each prophylactic therapy in the model *(bottom right)*. *NSAID*, Nonsteroidal anti-inflammatory drug; *PD*, pancreatic duct; *PEP*, post-ERCP pancreatitis.

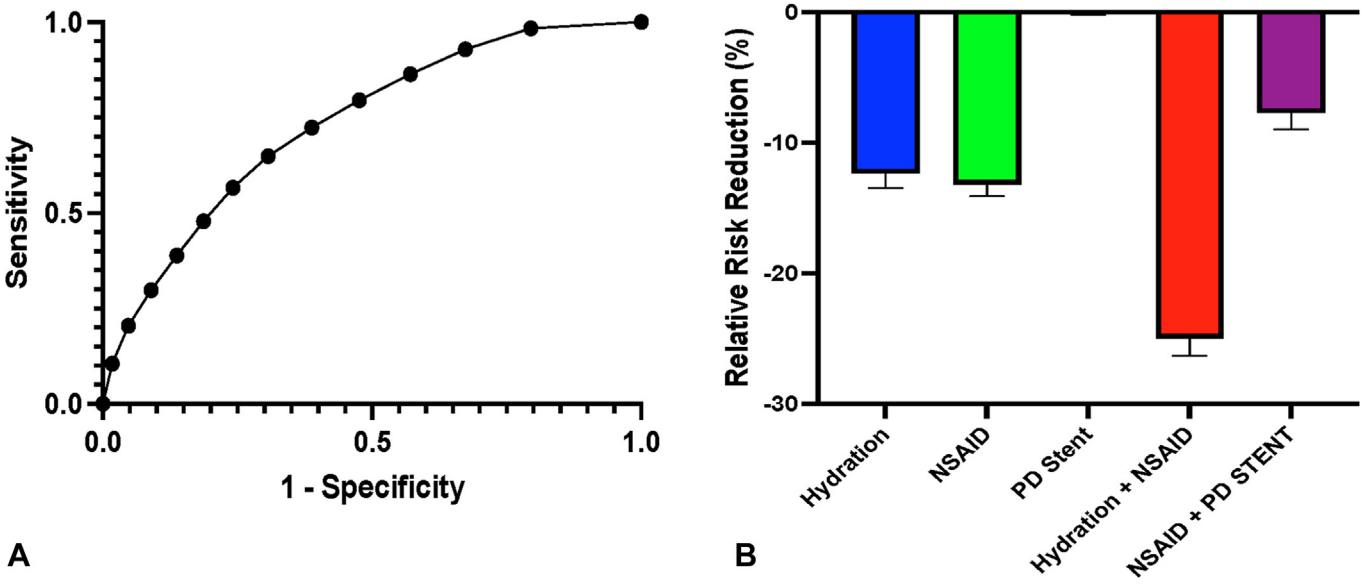



**Figure 3.** Validation of the PEP risk calculator in a prospective patient cohort (n = 135). **A,** The receiver operating characteristic curve for the PEP risk prediction model demonstrates an area under the curve of 0.74, outperforming the initial retrospective validation study, which had an area under the curve of 0.70. **B,** The predicted relative risk reduction under each prophylactic treatment paradigm calculated as the difference between the predicted risk of PEP for each therapy and the baseline predicted risk of PEP (ie, no intervention) divided by the baseline risk of PEP. *NSAID*, Nonsteroidal anti-inflammatory drug; *PD*, pancreatic duct; *PEP*, post-ERCP pancreatitis.

nonlinear interactions among variables. Freeman et al[11] demonstrated that the presence of multiple PEP risk factors had a synergistic effect on a patient's risk of developing PEP as well as on PEP severity, with marginal increases in PEP risk far larger than the individual odds ratios associated with those risk factors alone. Unlike linear regression, a machine learning model implicitly adjusts for the effects of interactions among risk factors on the probability of developing PEP. This is reflected in the fact that, although several risk factors, such as minor papillary sphincterotomy, carried a high incidence of PEP, they did not necessarily have the greatest impact on the predictive capacity of the model, given that other factors involved in the difficulty in cannulation and pancreatic trauma outweighed the contribution of minor papillary sphincterotomy.

Unlike prior studies, which have used either retrospective or prospective, nonrandomized, single-center cohorts of ERCP patients, this model was trained using patient-level data from large, multicenter, randomized controlled clinical trials, considered to be the reference standard for high-quality data for the development of artificial intelligence–based methods.[39] Although our model currently has an AUC of 0.70, we expect its predictive performance to improve significantly as more high-quality data are added to the model, as also noted during the validation of this model.

Furthermore, our data come from 12 separate randomized controlled trials, thereby encompassing a heterogeneous population of patients that serves to strengthen the predictive capabilities and generalizability of the model. Special emphasis was placed to ensure uniformity in the PEP prophylaxis regimens included based on our prior research.[23] Seven of the included 12 trials enrolled patients thought to be at moderate-to-high risk of developing PEP. Because our model predicts the risk of PEP and estimates treatment effects using data collected from these patients, it may be less accurate for lower-risk patients. We also recognize that the included studies used a wide range of protocols for aggressive hydration, which may affect the accuracy of risk reduction predictions for this prophylactic strategy and its derivatives.

Future studies using this model will focus on strengthening the model's predictive characteristics and validating it for use as a clinical decision-making tool integrated into the endoscopy workflow. Although the current web-based calculator has proven easy to use, this does require the endoscopy staff to access a separate system. Future work aims to integrate the calculator into the endoscopy reporting system so that most patient and procedural variables can be collected automatically for easier workflow integration in a busy endoscopy practice, with recommendations provided instantly to the endoscopist. As more randomized controlled trials on PEP risk and prevention are performed, we hope to train the model on additional datasets using the risk factors and treatments described, as well as studies containing traditionally understudied risk factors or therapeutic interventions currently under study, such as tacrolimus and protease inhibitors.[40,41] The collection of increasingly granular detail in these studies, such as combinations of prophylaxes, will also be useful in providing estimates or risk reduction for increasingly specific PEP prophylaxis regimens. Another avenue for future research will be to design and implement a prospective clinical trial to validate the use of the GBM model as a predictor of PEP, examine the treatment effects of the studied interventions, and assess whether the use of such a tool as a patient-specific, point-of-care decision-making aid confers any morbidity or mortality benefit for patients undergoing ERCP.

## CONCLUSIONS

We have developed a novel machine learning–based model for patient-specific PEP risk estimation capable of aiding clinicians in PEP prophylaxis selection and postprocedural monitoring. We demonstrate that this model is easy to use and performs with a high negative predictive value to identify patients at low risk for PEP who can safely receive limited PEP prophylaxis and be discharged home. Future studies will aim to further refine this machine learning model as additional clinical trial data are added and integrated into endoscopy reporting systems.

## DISCLOSURE

The following authors disclosed financial relationships: A. Kuo: Equity holder in Apollo Endosurgery and patent applications for intellectual property related to cancer early detection. S. S. Mok: Consultant for Sovereign Laboratories, ConMed, and Medtronic-Covidien Corporation. M. A. Khashab: Consultant for Boston Scientific and Olympus and royalties from Elsevier and UpToDate. E. J. M. van Geenen: Grants from Mylan and Olympus and personal fees outside the submitted work from MTW-Endoskopie. V. K. Singh: Grant support and personal fees outside the submitted work from AbbVie; advisory board participant for Ariel Precision Medicine, and Envara; grants from Orgenesis and Theraly; and advisory board participant for and stock options from Kyttaro. C. Tomasetti: Scientific advisory board member for PrognomiQ, Inc; royalty distributions from Exact Sciences (under a license agreement between Exact Sciences Corp and the Johns Hopkins University, C. Tomasetti and the university are entitled to royalty distributions); paid consultant for Bayer A; and patent applications for intellectual property related to cancer early detection (these arrangements have been reviewed and approved by the Johns Hopkins University in accordance with its conflict-of-interest policies). V. S. Akshintala: Cofounder and chief medical officer for Origin Endoscopy and Solv Endotherapy and consultant for Dragonfly Endoscopy and Olympus. All other

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*Abbreviations: AUC, area under the curve; GBM, gradient-boosted machine; NSAID, nonsteroidal anti-inflammatory drug; PD, pancreatic duct; PEP, post-ERCP pancreatitis.*

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